

Shobhit Singh



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EDUCATION

Iowa State University, College of Engineering Ames, Iowa
Bachelor + Master of Science in Computer Engineering

Fall 2024 – Fall 2027

Bennett University Gr. Noida, India (**Transferred to ISU**)
Bachelor of Technology in Computer Science Engineering

Fall 2022 – Spring 2024

TECHNICAL SKILLS

Embedded/Firmware: C/C++, Embedded C, ESP-IDF, Arduino, FreeRTOS, ESP32-S3/XIAO, UART/I2C/SPI/I2S, BLE/Wi-Fi, PWM/LED, DMA, PSRAM

Edge AI & CV: ESP- ESP-DL, TFLite Micro basics, OpenCV basics, camera pipelines, gesture/face detection, sensor fusion, fault-tolerant FSMs

Software/Data: Python, Java, JavaScript, HTML/CSS, NumPy, pandas, Streamlit, Android app development, HTTP APIs, Matplotlib

Security/Research: Cryptography, quantum-safe encryption, trusted execution environments, CAN/OBD-II basics, Qiskit, Linux, Git, VS Code

PROJECTS

[Salt.E](#) | Autonomous Salt Spreading Robot | CPRE 4910–4920 Senior Design

- Lead ESP32-S3 vision subsystem and Android companion app in a 6-person autonomous salt spreading rover that detects obstacles, coordinates cross-controller messaging, and supports precise salt dispensing.
- Implemented camera interface, DMA capture, CV obstacle-detection heuristics, app telemetry/control flows, and software synchronization for field testing.

[MIRA](#) & [MIRA 2.0](#) | Fault-Tolerant Multimodal Presence Verification | CPRE 5450 & CPRE/HCI 5750

- Built assistive robot with on-device face and gesture recognition.
- Integrated vision, audio, tracking, and companion software.
- Designed radar-camera activation controller for privacy-aware sensing.
- Built FreeRTOS tasks, radar parsing, and fail-safe logic.

Autonomous Wi-Fi Optimization & Router Placement Robot | CPRE 288

- Mapped Wi-Fi strength across rooms and visualized placement suggestions via Python GUI.

RESEARCH

Chaos Theory for Image Encryption

- Developed a novel chaos function for image encryption.

Advances in Trusted Execution Environments for Blockchain Security

- Reviewed TEEs for securing Hyperledger Fabric, with insights into Web3 and blockchain security.

Comparative Analysis of Pre- & Post-Quantum Encryption Algorithms

- Researched and compared encryption algorithms in the context of quantum-safe cryptography.

LEADERSHIP

General Secretary: *Tau Sigma Transfer Honors Society (ISU)*

Fall 2025 – Spring 2026

General Secretary: *ACM Student Chapter (Bennett Univ.)*

Aug 2023 – May 2024

Chairperson: *Quantum Computing Club (Bennett Univ.)*

Aug 2023 – May 2024

EXPERIENCE

Horse's Mouth Noida, India

Mar 2023 – Aug 2023

Operations and Technical Content Intern

- Partnered directly with CEO on mentor recruitment, business strategy, and workflow redesign, boosting website traffic **6x** and increasing user registrations from ~10 in 4 months to **10+ per month**.
- **Led a 6-member content team**, launching AI-driven SEO campaigns and producing high-engagement posts across Quora, Reddit, Twitter, and Instagram.